

## Why configure Kno for your use case?

Kno's AI engine parameters allow fine-tuning of the assistant's behavior based on the nature of your documents. This catalogue presents 4 preconfigured profiles, a hardware performance section, and a detailed glossary of each parameter.

## Configuration profiles by use case

### 01 Legal Analysis & Contracts

Dense PDF and DOCX files — clauses, case law, litigation

| Parameter              | Value      | Impact & Recommendation                                                                                     |
|------------------------|------------|-------------------------------------------------------------------------------------------------------------|
| Retrieved excerpts (K) | 5 – 7      | Cast wide to avoid missing any related clause. E.g. a termination clause may reference 3 separate articles. |
| Chunk size             | 800 – 1000 | Preserves entire coherent paragraphs. Avoids splitting a clause mid-sentence.                               |
| Overlap                | 200        | Essential: a legal clause split between two excerpts can reverse the meaning of an obligation.              |
| Temperature            | 0.0 – 0.1  | Strictly factual. No "creativity" tolerated on contractual data.                                            |

Context warning:  $K=7 \times \text{chunk}=1000 \sim 7,000$  tokens. With Mistral-7B (ctx 4096), prefer  $K=5$  if  $\text{chunk} > 900$ .

Target formats: PDF (contracts, deeds), DOCX (procedures, templates), TXT (exported case law).

### 02 Tabular Data & Numeric Reports

XLSX, CSV and tables embedded in PDFs

| Parameter              | Value  | Impact & Recommendation                                                                       |
|------------------------|--------|-----------------------------------------------------------------------------------------------|
| Retrieved excerpts (K) | 8 – 10 | Data is often spread across multiple rows/columns: cast wide to reconstruct the full context. |
| Chunk size             | 400    | Very short to precisely isolate a row or cell. A larger chunk buries the target value.        |
| Overlap                | 50     | Low: tabular data is independent row by row. High overlap would create noise.                 |
| Temperature            | 0.0    | Absolute mathematical precision. No paraphrasing accepted on numeric data.                    |

Performance note:  $K=8-10$  is compute-intensive. GPU recommended (RTX 3060+) for this profile. On CPU only, reduce to  $K=6$ .

Target formats: XLSX, CSV imported as TXT, PDF with tables. Prefer CSV export for purely numeric data.

### 03 HR Knowledge Base & Onboarding

Internal procedures, employee handbooks, DOCX and MD

| Parameter              | Value     | Impact & Recommendation                                                                           |
|------------------------|-----------|---------------------------------------------------------------------------------------------------|
| Retrieved excerpts (K) | 3         | Sufficient for a direct answer on well-structured documents. Avoids overly long responses.        |
| Chunk size             | 600       | Standard balanced format for smooth reading. Suited to procedure paragraphs and job descriptions. |
| Overlap                | 100       | Default recommended value. Sufficient to preserve context between two procedure steps.            |
| Temperature            | 0.4 – 0.5 | Makes the assistant's tone more human and welcoming for new employees.                            |

Target formats: DOCX (procedures, job descriptions), MD (internal wiki), TXT (onboarding notes).

Tip: create a dedicated "HR" collection to isolate this corpus from legal or technical documents.

## 04 Technical Documentation & IT Wikis

User guides, runbooks, IT support — many short files

| Parameter              | Value | Impact & Recommendation                                                                           |
|------------------------|-------|---------------------------------------------------------------------------------------------------|
| Retrieved excerpts (K) | 4     | Balance between speed and coverage. Technical docs are usually well structured and non-redundant. |
| Chunk size             | 500   | Targets a precise step in a procedure. Short enough to isolate a specific command or parameter.   |
| Overlap                | 150   | Preserves context from previous/next step. Crucial when steps are numbered and sequential.        |
| Temperature            | 0.2   | Balance between technical precision and clarity of explanation.                                   |

Target formats: MD (wikis, README), TXT (logs, runbooks), PDF (manufacturer manuals).

Tip: segment by product or system (e.g. collection "Network", "Servers", "Applications").

## Quick Reference Summary

| Profile           | K      | Chunk      | Overlap | Temperature | GPU?                 |
|-------------------|--------|------------|---------|-------------|----------------------|
| Legal & Contracts | 5 – 7  | 800 – 1000 | 200     | 0.0 – 0.1   | Recommended          |
| Tabular & Numeric | 8 – 10 | 400        | 50      | 0.0         | Strongly recommended |
| HR & Onboarding   | 3      | 600        | 100     | 0.4 – 0.5   | Optional             |
| Tech & IT Wiki    | 4      | 500        | 150     | 0.2         | Optional             |

## Performance & Hardware Recommendations

| Configuration         | GPU                        | Estimated response time | Compatible profiles  |
|-----------------------|----------------------------|-------------------------|----------------------|
| CPU only (default)    | None                       | 60 – 180 s              | HR, Tech (low K)     |
| CPU + entry-level GPU | RTX 3060 12 GB — layers 35 | 10 – 25 s               | All profiles         |
| Mid-range GPU         | RTX 4070 12 GB — layers 40 | 5 – 10 s                | All profiles, high K |

| Configuration       | GPU                        | Estimated response time | Compatible profiles  |
|---------------------|----------------------------|-------------------------|----------------------|
| High-end GPU        | RTX 4080/4090 — layers 50+ | 2 – 5 s                 | All + Mixtral 8x7B   |
| Apple Silicon (Mac) | M4 Pro / M5 Max            | 3 – 6 s                 | All — ideal law firm |

GPU setup: set `n_gpu_layers` in Admin → LLM Engine. Requires `llama-cpp` compiled with `CUDA` (Windows) or `Metal` (macOS).

## Kno Parameter Glossary

Detailed definitions for administrators and technical leads.

|                |                                                                                                                                                                                                                                                                                                                                                                              |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Retrieval      | <b>K — Retrieved excerpts</b> <p>Number of document fragments (chunks) retrieved from the vector index before generating a response. Higher K gives the AI more information but slows down the response.</p> <p><i>Example: K=3 reads the 3 closest passages to the question. K=7 covers more related clauses in a dense contract.</i></p> <p>Range: 1 – 10   Default: 3</p> |
| Indexation     | <b>Chunk size</b> <p>During indexing, Kno splits each document into small fragments called "chunks". This parameter defines the maximum size of each fragment in characters.</p> <p><i>Example: Chunk=400 isolates a table row. Chunk=1000 preserves the context of an entire legal paragraph.</i></p> <p>Range: 200 – 2000   Default: 700</p>                               |
| Indexation     | <b>Overlap</b> <p>Number of shared characters between two consecutive chunks. Prevents losing information at the boundary between two fragments.</p> <p><i>Example: Overlap=200 repeats the last 200 chars of chunk 1 at the start of chunk 2 — useful for a clause spanning two pages.</i></p> <p>Range: 0 – 500   Default: 100</p>                                         |
| LLM Generation | <b>Temperature</b> <p>Controls the AI model's "creativity level". Low value = deterministic, factual answers. High value = more varied and natural responses.</p> <p><i>Example: 0.0 = identical response to identical question (ideal for figures). 0.5 = varied phrasing (suited for HR).</i></p> <p>Range: 0.0 – 1.0   Default: 0.1</p>                                   |
| Performance    | <b>n_gpu_layers — GPU Layers</b> <p>Number of LLM model layers offloaded to the GPU (CUDA/Metal). Each layer moved from CPU to GPU significantly accelerates inference.</p> <p><i>Example: 0 = CPU only ~2-5 tok/s. layers=35 on RTX 3060 ~20-40 tok/s. layers=50 on RTX 4090 ~60-80 tok/s.</i></p> <p>Range: 0 – 100+   Default: 0 (CPU)</p>                                |
| File format    | <b>GGUF — Model format</b> <p>Standard file format for quantized (compressed) language models. Used by <code>llama.cpp</code>, the AI engine embedded in Kno. Self-contained files with no dependencies.</p>                                                                                                                                                                 |

*Example: Mistral-7B-Instruct-v0.3.Q4\_K\_M.gguf — 7B parameter model, 4-bit quantization medium quality, size 4.4 GB.*

**Extension:** .gguf | **Path:** models/

#### Architecture

### RAG — Retrieval-Augmented Generation

Architecture combining document base search (retrieval) with LLM response generation. Kno uses FAISS for vector search and llama.cpp for generation.

*Example: "What is the notice period in the Smith contract?" → (1) search K closest chunks, (2) send to LLM, (3) LLM cites source [1].*

**N/A — core architectural principle of Kno**

#### Search engine

### Vector index (FAISS)

Database storing documents as numerical vectors. Semantic search understands meaning: "terminate a lease" and "rental agreement cancellation" return the same results.

*Example: After indexing 90 PDFs → ~6,000 FAISS vectors. A question triggers a search among these 6,000 vectors in milliseconds.*

**Path:** vectorstore/ — 1 index per collection

#### LLM Behavior

### System prompt

Hidden instruction sent to the model before each question, defining its role and constraints. In Kno, enforces using only provided excerpts and citing sources.

*Example: Customize to specify "You are a corporate law specialist assistant. If uncertain, say so explicitly."*

**Free text field — Admin > LLM Engine**

#### Organization

### Collection

Isolated storage space with its own documents and FAISS index. Kno searches only in the selected collection when answering a question.

*Example: Collection "Contracts" (200 client contracts) + Collection "GDPR" (policies). A question in "Contracts" never searches "GDPR".*

**Unlimited count — Admin > Collections**